

SQL Important Functions – Placement & LeetCode Cheat Sheet

Focus on these functions and clauses first. They cover a large portion of common SQL interview and LeetCode problems.

1. Aggregate Functions — MOST IMPORTANT

Function	Use	Example
COUNT()	Count rows / non-NULL values	COUNT(*)
SUM()	Total	SUM(amount)
AVG()	Average	AVG(salary)
MAX()	Maximum	MAX(salary)
MIN()	Minimum	MIN(salary)

2. Conditional Logic

Function	Use	Example
CASE WHEN	Conditional calculation	CASE WHEN state='approved' THEN amount ELSE 0 END
IF()	MySQL conditional	IF(state='approved', amount, 0)

3. NULL Handling

Function	Use	Example
COALESCE()	First non-NULL value	COALESCE(value, 0)
IFNULL()	Replace NULL	IFNULL(value, 0)
NULLIF()	Return NULL when two values are equal	NULLIF(a, b)

4. String Functions

Function	Use	Example
CONCAT()	Join strings	CONCAT(first_name, ' ', last_name)
LOWER()	Lowercase	LOWER(name)
UPPER()	Uppercase	UPPER(name)
LENGTH()	String length	LENGTH(name)
LEFT()	Characters from left	LEFT(name, 3)
RIGHT()	Characters from right	RIGHT(name, 3)
SUBSTRING()	Extract part of string	SUBSTRING(name, 1, 3)
TRIM()	Remove surrounding spaces	TRIM(name)
REPLACE()	Replace text	REPLACE(name, 'A', 'B')

5. Date Functions — VERY IMPORTANT

Function	Use	Example
YEAR()	Get year	YEAR(order_date)
MONTH()	Get month number	MONTH(order_date)
DAY()	Get day	DAY(order_date)
DATE()	Get date portion	DATE(timestamp_col)
DATE_FORMAT()	Format date	DATE_FORMAT(trans_date, '%Y-%m')
DATEDIFF()	Difference in days	DATEDIFF(end_date, start_date)
TIMESTAMPDIFF()	Difference in chosen unit	TIMESTAMPDIFF(YEAR, start, end)

6. DATE_FORMAT() — Memorize These

Specifier	Meaning	Example
%Y	4-digit year	2026
%y	2-digit year	26
%m	Month 01–12	09
%M	Full month name	September
%d	Day 01–31	05
%H	Hour 00–23	14
%i	Minute 00–59	30
%s	Second 00–59	45
%W	Full weekday name	Monday

7. Window Functions — IMPORTANT FOR INTERVIEWS

Function	Use	Example
ROW_NUMBER()	Unique row numbering	ROW_NUMBER() OVER (...)
RANK()	Ranking with gaps	RANK() OVER (ORDER BY salary DESC)
DENSE_RANK()	Ranking without gaps	DENSE_RANK() OVER (...)
LAG()	Previous row value	LAG(salary) OVER (...)
LEAD()	Next row value	LEAD(salary) OVER (...)
SUM() OVER()	Running/window total	SUM(amount) OVER (...)
AVG() OVER()	Window average	AVG(salary) OVER (...)

8. Essential Clauses — MORE IMPORTANT THAN MEMORIZING FUNCTIONS

Clause	Purpose
SELECT	Choose columns/calculations

Clause	Purpose
FROM	Choose table
JOIN	Combine tables
LEFT JOIN	Keep all rows from left table
WHERE	Filter rows before grouping
GROUP BY	Create groups for aggregation
HAVING	Filter groups after aggregation
ORDER BY	Sort results
LIMIT	Restrict number of rows
DISTINCT	Remove duplicate results

9. Operators / Conditions

Operator	Use
=	Equal
<> / !=	Not equal
>, <, >=, <=	Comparison
BETWEEN	Range
IN	Match a list
LIKE	Pattern matching
IS NULL	Check NULL
IS NOT NULL	Check non-NULL
AND / OR / NOT	Combine conditions

10. SQL Patterns You Should Know

Pattern	Think About
Top / highest	ORDER BY ... DESC + LIMIT
Count by category	GROUP BY + COUNT()
Conditional count	COUNT(CASE WHEN ... THEN 1 END)
Conditional total	SUM(CASE WHEN ... THEN amount ELSE 0 END)
Average with no value	COALESCE(AVG(...), 0)
Monthly result	DATE_FORMAT(date, '%Y-%m') + GROUP BY
Previous row	LAG() OVER (...)
Ranking	RANK() / DENSE_RANK() OVER (...)
Find duplicates	GROUP BY + HAVING COUNT(*) > 1
Compare rows	LAG()/LEAD() or self JOIN

Quick Revision Order

1. SELECT, WHERE, GROUP BY, HAVING, ORDER BY
2. COUNT, SUM, AVG, MAX, MIN
3. JOIN and LEFT JOIN
4. CASE WHEN, IF
5. COALESCE / NULL handling
6. DATE_FORMAT, YEAR, MONTH, DATEDIFF
7. String functions
8. Window functions: ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD

Placement tip: Don't just memorize functions. Practice identifying whether a problem needs filtering, joining, grouping, aggregation, conditional logic, date manipulation, or a window function.